Enhancing Customer Experience Through Real-Time Multimodal Speech Emotion Recognition

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***Abstract* - Customer service satisfaction relies heavily on recognizing caller emotions in real time. However, standard audio-based Speech Emotion Recognition (SER) models struggle with the "Valence Gap," while text-capable Transformer models are typically too large and slow for real-time applications. To address these challenges, we propose a lightweight, multimodal SER architecture designed specifically for low-latency customer service environments. Our system bridges the valence gap by combining acoustic features with lexical information. It extracts high-level acoustic features via a compressed WavLM-Small Transformer and captures lexical context using Whisper ASR and DistilBERT token embeddings. These are fused with low-level acoustic properties through a Multi-Level Co-Attention mechanism. We used lightweight multimodal architecture trained on the IEMOCAP dataset supplemented by robust data augmentation, which is then stress tested against a Call Center corpus. The pre-trained model achieved a mean Weighted F1-score of 0.715 on the IEMOCAP dataset. Crucially, the pipeline demonstrated an end-to-end latency of 508.3 ms, providing an efficient and lightweight model well within the 2000 ms threshold required for real-time deployment, while still having sentiment depth for accurate classifications.**

Keywords—Speech Emotion Recognition (SER), Multimodal, Real-Time, Customer Service, Deep Learning, Transformer, Low Latency

I. INTRODUCTION

Customer service satisfaction relies heavily on positive emotional experiences, making real-time emotion detection vital for business operations [1]. With a high volume of calls [2] and AI-automated interactions projected to increase by 2026 [3], Speech Emotion Recognition (SER) has emerged as a crucial HCI tool [4]–[6], enabling AI to understand caller emotional contexts beyond mere transcription.

Despite advances in Deep Learning for SER [7], [8], applying it to customer service presents challenges related to environmental noise and real-time latency. Traditional audio-only models struggle with the "Valence Gap," failing to distinguish between emotions with similar acoustic energy without linguistic context [9]. While large-scale, self-supervised Transformers such as Wav2Vec 2.0 and HuBERT can extract linguistic information directly from raw audio to close this gap [9], [10], their large parameter sizes make them too slow for low-latency inference on resource-constrained devices [11].

To address these constraints, this study proposes a lightweight, multimodal SER architecture tailored for business customer service. We combined audio and text inferences to effectively bridge the Valence Gap without heavy computation. A compressed Transformer model [11] is used to extract implicit linguistic meaning, and its high-level features are then fused with low-level acoustic properties (MFCCs/Spectrogram) via a Multi-Level Co-Attention mechanism [12]. Simultaneously, explicit lexical information is analyzed using a Multi-scale CNN (MSCNN), creating a robust multimodal system that maintains the low latency required for real-time calls. The primary goal of this research is to create a low-latency multimodal SER model that balances real-time efficiency and semantic depth to bridge the Valence Gap.

II. RELATED WORK

A multimodal SER incorporating dialogue context was proposed in [13], combining a contextual transformer, a cross-modal transformer for extracting MFCCs and text, and a Graph Convolutional Network (GCN) for fusion, achieving 83.74% Weighted Accuracy (WA) and 84.27% Unweighted Accuracy (UA) on IEMOCAP transcripts. Building on multimodal fusion, [14] proposed a Multimodal Cross- and Self-Attention Network (MCSAN), featuring an inter-modal module that lets acoustic and lexical features acknowledge and update each other, in parallel with an intra-modal module that processes each modality's details separately.

A study in [15] proposed an MSCNN-SPU-ATT model, which processes audio and text in parallel via MSCNN, represents acoustic features using Statistical Pooling Units (SPUs), and fuses both modalities through bi-modal attention, achieving 80.3% WA and 81.4% UA. Similarly, [16] developed a Key-Sparse Transformer (KS-Transformer) that identifies informative speech segments and drops uninformative features, achieving 74.3% WA and 75.3% UA on IEMOCAP.

Self-supervised features were also explored in [10] using an Upstream + Downstream architecture (Wav2Vec 2.0, HuBERT, Mean Average Pooling, and ECAPA-TDNN), achieving 77.36% WA and 77.76% UA on IEMOCAP. Addressing spectrogram-based limitations, [17] proposed an Attentive Time-Frequency Neural Network (ATFNN) modeling correlations across frequency bands and time frames, outperforming state-of-the-art models with 73.81%

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WA and 64.48% UA. To reduce model size, [11] developed Vesper, a compact task-specific Pre-Trained Model that compresses Transformer layers via direct parameter extraction/averaging and RMS energy (RMSE); its four-layer variant, Vesper-4, achieved 68.4% WA on IEMOCAP, outperforming larger PTMs such as WavLM despite being only 67.1% of its size.

Robustness to real-world audio conditions has also been addressed. [18] tackled discrepancies between clean and noisy audio using Coherent-to-Diffuse Power Ratio (CDR) and Weighted-Prediction Error (WPE) dereverberation, Iterative Distorted Feature Cut (IDFC), F-score feature selection, and synthetic Room Impulse Response injection, achieving classification accuracy between 79.44% and 95.89%. [19] augmented training data with white noise and spectrogram shifting across three models MLP, CNN, and CNN+BiLSTM with CNN+BiLSTM proving most robust, reaching 100% accuracy on TESS, 99.50% on EmoDB, and 90.12% on RAVDESS.

Finally, [20] developed a multilingual real-time SER extracting low-level features (ZCR, RMSE, MFCC) processed via a 1D CNN with LSTM and attention, which outperformed a 2D CNN baseline with accuracies of 96.72% on EMO-DB, 97.13% on SAVEE, 96.72% on ANAD, and 88.39% on BAVED.

A review of previous works reveals a common pattern of using Transformers, CNNs, LSTMs, and attention mechanisms to extract emotional features from audio, with several studies improving robustness through data augmentation or feature selection. However, applying these models in real-time scenarios still presents common issues, such as heavy computation, susceptibility to latency, and limited semantic depth. The proposed model aims to bridge this gap by combining noise handling with a lightweight multimodal attention model, reducing latency and computational cost while improving emotional understanding in a real-time system.

III. METHODOLOGY

This section describes the pipeline of the proposed model. As shown in Fig. 1, the model development followed these steps: data preparation, data augmentation, feature extraction, model development, model implementation, and performance evaluation

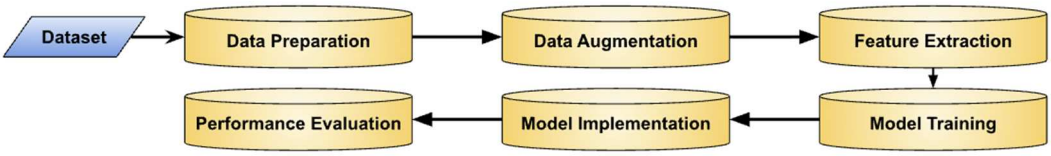


Fig. 1. Model development pipeline

A. Dataset

This study uses IEMOCAP, a widely recognized SER benchmark providing strictly annotated audiovisual recordings crucial for mapping basic emotions [7], as the primary dataset for training. The Customer Call Center Dataset, sourced from Kaggle, is then used purely as a cross-domain stress test to evaluate the model's capabilities under real-world telecommunication conditions. The combination of these two datasets is detailed in Table 1.

Table 1.Table of dataset details

Property	Customer Call Center Dataset	IEMOCAP
Source	Kaggle: https://www.kaggle.com/datasets/rafaqatkhan608/customer-call-center-dataset-analysis	Kaggle: https://www.kaggle.com/datasets/sangayb/iemocap
Type	Simulated Call Center	Scripted & Improvised Dydalic
Modalities	Audio (.wav) + Text Transcripts	Audio + Text + Motion Captures
Emotion Labels	Angry, Happy, Neutral, Frustrated, Confused	Anger, Happiness, Sadness, Frustration, Neutral
Duration	Variable Clip Lengths	12 Hours
Speakers	Multiple Simulated Customers	10 Actors
Primary Role	Real-World Domain Stress Test	Benchmark and Generalization

B. Data Preparation

To ensure that the raw data meet the input requirements of deep representation learning models [8], a rigorous data preparation workflow is applied to both the audio and text modalities, as shown in Fig. 2. First, all audio recordings are converted to mono format and standardized to a 16 kHz sampling rate. Next, a Voice Activity Detection (VAD) algorithm is used to identify and trim silent intervals at the beginning, end, and within the speech [4]. The continuous audio signal is then segmented into uniform 4-second frames to maintain consistent tensor dimensions during the training phase.

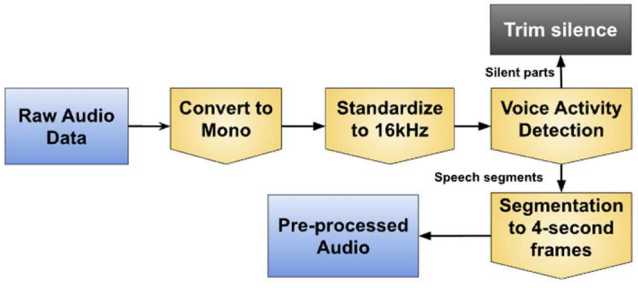


Fig. 2. Audio pre-processing pipeline

C. Data Augmentation

To improve generalization robustness in noisy real-time scenarios, we systematically apply three data augmentation strategies to the clean audio signals in the IEMOCAP dataset.

1) White Noise Addition

To simulate background sound, Gaussian white noise is randomly inserted into the clean audio signal during training [19]. The Signal-to-Noise Ratio (SNR) is dynamically varied between 5 dB and 25 dB to train the model's ability to distinguish features of emotional speech from static background noise.

2) Spectrogram Shifting

To mimic network latency and packet loss, shifting and masking techniques are applied to the time and frequency domains of the spectrogram representation. This forces the model to classify emotions accurately even when relying solely on partial or abruptly truncated acoustic features [17].

3) Synthetic Room Impulse Response

To address echo, the audio signal is convolved with a Synthetic Room Impulse Response (SRIR) [18]. This procedure simulates acoustic reflection within a room.

D. Feature Extraction

Feature extraction is at the core of the proposed architecture, where information is extracted from two different modalities to comprehensively capture both the physical prosodic elements and the semantic meaning, so that it can effectively bridge the valence gap.

1) Low-Level Acoustic Features

These features are extracted directly from the audio signal to capture the physical characteristics and basic intonation profile of the caller's voice.

a) MFCC (+ Delta + Delta-Delta)

Mel-Frequency Cepstral Coefficients (MFCCs) are utilized because of their ability to model human auditory perception. A total of 40 MFCCs are extracted. This is crucial for detecting abrupt emotional transitions, such as a sudden shift from neutral to angry [7].

b) Mel-Spectrogram

This representation transforms the 1D audio signal into a 2D spectral image based on the Mel scale. The Mel-Spectrogram preserves the spatial information of frequency over time, enabling the model to capture local time-frequency patterns that are often lost in 1D feature extraction [17].

c) Statistical Functionals (ZCR, Pitch, RMS mean, std, skew, kurtosis)

Frame-level features such as the Zero Crossing Rate (ZCR) to distinguish unvoiced background noise from voiced human speech, Pitch (F0) to measure the vocal fundamental frequency, and Root Mean Square (RMS) to quantify volume intensity, are extracted and aggregated. Four statistical moments: mean, standard deviation, skewness, and kurtosis are computed from these features to quantify the volatility of the caller's emotion [4].

2) High-Level Acoustic Features

Unlike low-level features that measure physical properties, high-level features are automatically learned by a deep neural network to understand the context and implicit linguistic aspects of the waveform without explicit transcription. To extract this, the raw waveforms are processed using WavLM-Small, a self-supervised Transformer model. Extraction is performed by taking representation vectors from the hidden states of the final encoder layers. This yields context-rich acoustic representations with minimal latency, making them highly suitable for real-time operational constraints.

3) Lexical Information

The textual modality explicitly extracts semantic meaning to ensure that the system can understand the emotionally charged words or phrases spoken by customers.

a) Whisper ASR transcription

The audio signal is transcribed into text using the Whisper Automatic Speech Recognition (ASR) model. Since the datasets already include the transcriptions of the audio files, the Whisper ASR is applied only during real-time deployment testing.

b) DistilBERT token-level embeddings

The transcribed text is subsequently tokenized and processed using DistilBERT. The deployment of this distilled model ensures highly accelerated Natural Language Processing (NLP) without significantly compromising contextual sentence comprehension.

E. Model Architecture & Training

Fig. 3 shows the proposed model architecture as a three-branch multimodal system, where each branch processes one of the three extracted feature types: low-level acoustic features, high-level acoustic features, and lexical information. Table 2 specifies the input and output dimensions of each component vector.

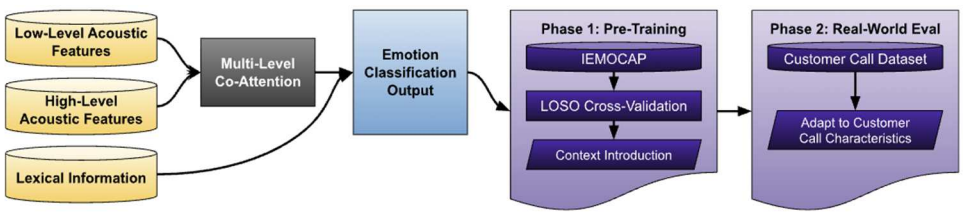


Fig. 3. Two-phase training pipeline

The first two branches, which process acoustic features, are fused through a Multi-Level Co-Attention system where hidden states of the features are projected to share a dimensional space (1) and (2), before doing cross-attention to align the features and finalize the acoustic vector (3)-(5).

The co-attention module operates through three levels with four attention heads (H=4). We denote the hidden states of WavLM as $T \in R^{T1\times256}$ and MFCC sequence $A \in R^{T2\times128}$. Both are projected into 256-d before fusion:

$$T' = W_T H_{WavLM} \quad (1)$$

$$A' = W_A H_{MFCC} \quad (2)$$

where $W_T \in R^{768\times256}$ and $W_A \in R^{40\times128}$.

The three level co-attention is denoted as in (3)-(5):

$$L_1 = LN(T' + MHA(T', A', A')) \quad (3)$$

$$L_2 = LN(A' + MHA(A', T', T')) \quad (4)$$

$$L_3 = LN(C + MHA(C, C, C)) \quad (5)$$

where LN is the Layer Normalization and $C = [L_1 \| L_2]$. The output is a 256-dimension vector ($f_{acoustic} \in R^{256}$).

The MSCNN branch projects a 256-d vector with three parallel 1D convolutions. For k □ {2, 3, 5}, f_k is computed shown in (6) to then produce the lexical vector as in (7).

$$f_k = MaxPool(GELU(BN(Conv1D_k(E)))) \quad (6)$$

$$f_{lex} = W_{fuse}[f_2 \| f_3 \| f_5] \in R^{256} \quad (7)$$

where $E \in R^{64\times768}$.

The statistical branch computed over MFCC, ZCR, RMS, and Pitch produces a 738-dimension vector, projected into a 128-dimension vector as shown in (8).

$$f_{stat} = W_2 . GELU(W_1 . LN(s)) \in R^{128} \quad (8)$$

These branches then are concatenated, producing a 640-d vector shown in (9) that is then inserted into the MLP classifier as 5 classes as computed in (10).

$$f_{fused} = [f_{acoustic} \| f_{lex} \| f_{stat}] \in R^{640} \quad (9)$$

$$\hat{y} = Softmax(W_3 . GELU(W_2 . GELU(W_1 f_{fused}))) \quad (10)$$

Table 2. Model architecture table detailing input and output dimension of each component

Component	Input Dimension	Output Dimension
WavLM	768	256
MFCC	40	128
Co-Attention	256/128	256
MSCNN	64x768	256
Statistical	738	128
Fusion	256+256+128	640
MLP Classifier	640	5

F. Performance Evaluation

We used four metrics: WF1 to primarily measure the model's classification performance, Unweighted Average Recall (UAR) as the standard for SER evaluation, accuracy to evaluate the overall classification capability of the model, and end-to-end latency to test the real-time performance.

We also conducted an ablation study on our model where we bypassed four components one by one: the WavLM branch, the MSCNN branch, the low-level acoustic branch, and also we omitted the co-attention architecture.

IV. Results

A. IEMOCAP with LOSO Cross Validation

Results of the IEMOCAP testing with LOSO are shown as a confusion matrix for each session, shown in Fig. 4.

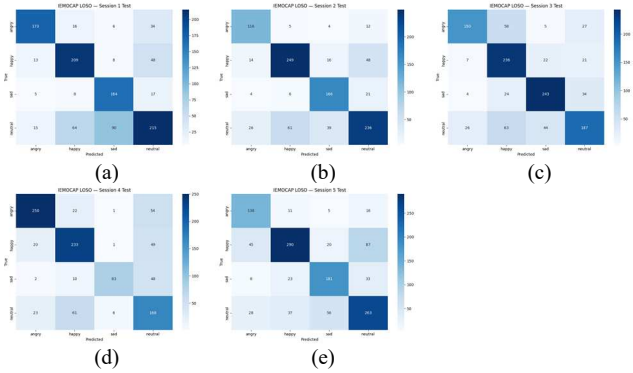


Fig. 4. Confusion matrix on all five LOSO sessions: (a) session 1, (b) session 2, (c) session 3, (d) session 4, and (e) session 5

Table 3 summarizes the weighted F1, UAR, and overall accuracy scores for each LOSO fold. The model achieved consistent results across all five sessions. A standard deviation of 0.018 on the weighted F1 and accuracy scores, and 0.028 on the UAR, indicates that the model does not overfit. These results are compared with previous work in Table 4 to assess the model's performance relative to other models on the IEMOCAP dataset. Although our model does not outperform previous models, it trades off a marginal amount of performance to achieve more fast latency.

Table 3. Evaluation results of IEMOCAP Dataset with LOSO cross validation

Session	Weighted F1	UAR	Accuracy	Best	Worst
Session 1	0.699	0.728	0.701	Sad (0.845)	Neutral (0.560)
Session 2	0.748	0.776	0.750	Angry (0.847)	Neutral (0.652)
Session 3	0.707	0.708	0.709	Happy (0.825)	Angry (0.625)
Session 4	0.717	0.691	0.712	Angry (0.764)	Sad (0.580)
Session 5	0.703	0.723	0.703	Angry (0.812)	Happy (0.656)
Mean ± Std	**0.715 ± 0.018**	**0.725 ± 0.028**	**0.715 ± 0.018**	-	-

Table 4. Model performance comparison on the IEMOCAP dataset to previous work

Model	Modality	WF1/WA (%)	UAR /UA (%)
Upstream + Downstream [10]	Audio	77.37	77.76
Cross-modal + GCN [13]	Audio + Text	83.74	84.27
MSCNN-SPU [15]	Audio + Text	80.3	81.4
KS-Transformer [16]	Audio + Text	74.3	75.3
ATFNN [17]	Audio	73.81	64.48
Proposed	**Audio + Text**	**71.5**	72.5

B. *Ablation Study Results*

Results of the architectural and modality ablation study, obtained after bypassing each component of the model are shown in Table 5, ranked from most to least significant.

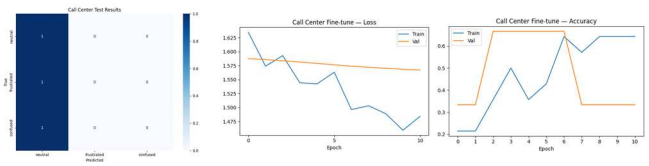
Table 5. Architectural and modality ablation study results

Component Omitted	Weighted F1	Accuracy
Baseline	**0.715**	**0.715**
Co-Attention Module	0.411 (-0.304)	0.411 (-0.304)
MSCNN Lexical Branch	0.465 (-0.250)	0.490 (-0.225)
Low-Level Acoustic Branch	0.536 (-0.179)	0.543 (-0.172)
WavLM Branch	0.666 (-0.049)	0.670 (-0.045)

Omitting the co-attention module results in the biggest drop of performance, dropping the performance of WF1 and Accuracy by 0.304, a 42.5% decrease, indicating that the fusion of branches contributed to a major part of the models performance. The lexical branch omitted resulted in a 0.25 decrease of WF1 and 0.225 of accuracy, which means the lexical information is still contributing a good portion of the performance. The low-level acoustic branch omitted dropped the WF1 by 0.179 and accuracy by 0.172, and the WavLM branch decreased by only 0.049 for WF1 and 0.045 for accuracy, indicating marginal effect to the performance.

C. *Cross-Domain Stress Test Using Call Center Corpus*

The cross-domain stress test was conducted using the Call Center corpus. This dataset had the limitation of containing only 20 labelled samples. Nevertheless, a 70/15/15 split was still applied during the stress test. Due to the small dataset, the testing results showed that all three test samples were predicted as neutral. Fig. 5 illustrates this instability, showing a marginal decrease in training loss from 1.63 to 1.47, while accuracy only briefly increased to 0.67 before falling back to 0.33.



(a) (b) (c)

Fig. 5. Call center corpus evaluation: (a) confusion matrix, (b) training loss, and (c) accuracy

Results show that the small sample size of the dataset negatively affected the stress test. A richer corpus is needed to perform cross-domain stress tests more effectively.

D. *Model Latency Testing*

End-to-end latency testing is done to assess if the model is suitable for real-time deployment. The latency testing was done on a standard hardware environment. We used hardware with a 10-core (4 performance cores, 6 efficiency cores) CPU and 16 GB of RAM.

The system achieved a mean end-to-end latency of ~508.3 ms, well within the 2000 ms limit for effective real-time use. Audio processing took the fastest time with only ~0.3 ms. MFCC, Mel-Spec, and Statistical Functionals branch only took ~37.2 ms. WavLM also took a significant amount of time with ~102.1 ms, contributing to around 20% of the total time. DistilBERT only took ~10.4 ms, and the Co-Attention, MSCNN, and MLP Classifier branch only took ~4.7 ms. Lastly, Whisper ASR consumed the most time with ~321.8 ms or 63.3% of the total time.

V. CONCLUSION

This study proposes a multimodal real-time SER approach combining a WavLM transformer, low-level acoustic features, and DistilBERT/MSCNN branches with a Co-Attention module, designed for deployment in real-world customer service scenarios. Using the IEMOCAP corpus, the model produced stable LOSO cross-validation results across all emotions, achieving a mean WF1 of 0.715 and UAR of 0.725. An ablation study further confirmed that the co-attention module and lexical branch are the most significant contributors to classification performance.

When stress tested on the call center corpus, the model's 20-sample dataset proved insufficient for reliable cross-domain evaluation, indicating the need for a larger corpus. Latency testing, however, confirmed the model's real-world viability, with a mean latency of approximately 508.3 ms well within the 2,000 ms threshold demonstrating that the model trades a marginal amount of accuracy for significantly lower latency.

These results point to two directions for future work: obtaining a larger customer call center corpus to refine cross-domain evaluation, and exploring more efficient ASR alternatives to further reduce pipeline latency and improve overall system lightweightness.

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AUTHOR CONTRIBUTION STATEMENT

Nathan Valliant: Methodology, Conceptualization, and Model Creation, **Muhammad Mirza:** Methodology, Conceptualization, and Evaluation, **Ivan Sebastian Edbert:** Validation, Supervision, and Project Administration, **Samson Ndruru:** Validation, Supervision, and Project Administration.

Data Availability

The data for this study is publicly available at the following links:

IEMOCAP Dataset: https://www.kaggle.com/datasets/sangayb/iemocap

Customer Call Dataset: https://www.kaggle.com/datasets/rafaqatkhan608/customer-call-center-dataset-analysis

AI Usage Declaration

We declare that we used AI tools such as Perplexity to help find related research and works, and Gemini to help prepare the manuscript for grammar correction. All AI-generated content was reviewed, edited, and validated by the authors to ensure accuracy, clarity, and adherence to academic standards.

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